Product Requirements Document (PRD)

Product Name: Retinal Scan Health AI (Working Title) **Document Version:** 1.0 **Date:** May 24, 2026

1. Executive Summary

This application leverages advanced artificial intelligence and smartphone camera technology to analyze retinal photos and detect potential indicators of systemic diseases, specifically diabetes, hypertension, and early-stage Alzheimer's disease. The goal is to provide accessible, non-invasive, and early preliminary screening capabilities directly to users.

2. Product Vision

To democratize early health screening by transforming everyday smartphones into powerful diagnostic support tools, empowering individuals to proactively monitor their systemic health through accessible retinal analysis.

3. Target Audience

- **Primary:** Individuals seeking proactive health monitoring, especially those with family histories of diabetes, hypertension, or Alzheimer's.
- **Secondary:** Healthcare professionals in remote or under-resourced settings looking for accessible triage tools (Note: Regulatory approval required for clinical use).
- **Tertiary:** Elderly care facilities or home health aides monitoring residents/patients.

4. Problem Statement

Early detection of systemic diseases like diabetes, hypertension, and Alzheimer's is crucial for effective management and improved outcomes. However, traditional screening methods often require specialized equipment, clinical visits, and can be inaccessible or expensive for many.

5. Proposed Solution

An Android application that guides a user to take a high-quality photo of their retina using their smartphone camera (potentially requiring a simple, low-cost macro lens attachment). An on-device or cloud-based AI model analyzes the vascular structure and biomarkers in the image to provide a risk assessment for the target diseases.

6. Key Features & Requirements

6.1. Core Functionality (MVP)

- **User Account & Onboarding:** Secure registration and comprehensive onboarding explaining the process, privacy, and limitations (it is a *screening tool*, not a *diagnosis*).
- **Image Capture Guide:** An interactive, AR-assisted guide to help users capture a clear, focused image of the retina. This is the most critical UX challenge. Features include:

    - Focus and stability assistance (e.g., gyroscope integration).
    - Auto-capture when optimal conditions are met.
- **Image Quality Assessment (IQA):** An immediate, lightweight model to verify if the captured image is of sufficient quality for analysis. If not, prompt the user to retake.
- **AI Analysis Module:** The core machine learning engine.
    - Analyzes the verified image for biomarkers associated with diabetes (e.g., microaneurysms), hypertension (e.g., arteriovenous nicking), and Alzheimer's (e.g., retinal thinning/vascular changes).
    - Requires robust, medically validated datasets for training (This is a major prerequisite).
- **Results Dashboard:** Presents the analysis clearly and simply.
    - Uses risk categories (e.g., Low, Moderate, High Risk) rather than definitive diagnoses.
    - Clear disclaimers emphasizing the need for professional medical consultation.
- **Data Security & Privacy:** End-to-end encryption, strict adherence to HIPAA/GDPR (or local equivalents) regarding health data storage and processing.

6.2. Future Enhancements (Post-MVP)

- **Integration with Electronic Health Records (EHR):** Allowing users to securely share results with their doctors.
- **Trend Tracking:** Analyzing changes in retinal images over time to track disease progression or intervention efficacy.
- **Expanded Disease Detection:** Adding models for other conditions (e.g., glaucoma, macular degeneration, cardiovascular risk).
- **Cloud Sync & Backup:** Secure cross-device access.

7. Technical Requirements

- **Platform:** Android (initially), targeting modern devices with high-resolution cameras.
- **Frontend Framework:** Kotlin (Native Android) or Flutter/React Native (if cross-platform is a near-term goal).
- **Backend/Cloud:** AWS, Google Cloud, or Azure for secure data processing (if not fully on-device) and user management.
- **AI/ML Frameworks:** TensorFlow Lite or PyTorch Mobile for on-device inference; larger frameworks if cloud-based.

8. Dependencies & Risks

- **Regulatory Approval:** The most significant hurdle. Claiming to detect diseases requires rigorous validation and clearance from bodies like the FDA (US) or EMA (Europe). The MVP must be positioned strictly as a "wellness" or "informational" tool until cleared.
- **Image Quality:** Smartphone cameras vary wildly. Achieving consistent retinal imaging without specialized attachments is technically difficult.

unnecessary anxiety; false negatives are dangerous. Access to large, annotated medical datasets is crucial.

- **Data Privacy:** Handling sensitive health data carries significant legal and reputational risks.

9. Success Metrics (KPIs)

- **User Acquisition & Retention:** Number of downloads, active users, and frequency of use.
- **Image Capture Success Rate:** Percentage of users who successfully capture a scannable image on the first or second attempt.
- **Diagnostic Concordance (Clinical Trials):** Agreement between the app's risk assessment and standard clinical diagnosis (crucial for regulatory approval).
- **App Store Ratings & User Feedback:** General user satisfaction and ease of use.